

Task 4

Question 1

A factory manufactures two products A and B on three machines X , Y , and Z . Product A requires 10 hours of machine X , 5 hours of machine Y , and 1 hour of machine Z . Product B requires 6 hours, 10 hours, and 2 hours of machines X , Y , and Z respectively. The profit contribution of products A and B are Rs. 23/- per unit and Rs. 32/- per unit respectively. In the coming planning period, the available capacity of machines X , Y , and Z are 2500 hours, 2000 hours, and 500 hours respectively. Find the optimal product mix for maximizing the profit.

Solution

Step 1: Define Decision Variables

Let:

- x_1 = Number of units of Product A to produce
- x_2 = Number of units of Product B to produce

Step 2: Formulate the Objective Function

The goal is to maximize profit. The profit contributions are:

- Rs. 23 per unit of Product A
- Rs. 32 per unit of Product B

The objective function is:

$$\text{Maximize } Z = 23x_1 + 32x_2$$

Step 3: Formulate the Constraints

The constraints are based on the available machine hours.

1. **Machine X :**

$$10x_1 + 6x_2 \leq 2500$$

2. **Machine Y :**

$$5x_1 + 10x_2 \leq 2000$$

3. **Machine Z :**

$$x_1 + 2x_2 \leq 500$$

4. **Non-Negativity Constraints:**

$$x_1 \geq 0, \quad x_2 \geq 0$$

Step 4: Convert Inequalities to Equations

Introduce slack variables s_1 , s_2 , and s_3 to convert the inequalities into equations:

1. **Machine X:**

$$10x_1 + 6x_2 + s_1 = 2500$$

2. **Machine Y:**

$$5x_1 + 10x_2 + s_2 = 2000$$

3. **Machine Z:**

$$x_1 + 2x_2 + s_3 = 500$$

4. **Non-Negativity Constraints:**

$$x_1 \geq 0, \quad x_2 \geq 0, \quad s_1 \geq 0, \quad s_2 \geq 0, \quad s_3 \geq 0$$

Step 5: Set Up the Initial Simplex Tableau

The initial Simplex tableau is constructed as follows:

	x_1	x_2	s_1	s_2	s_3	RHS
s_1	10	6	1	0	0	2500
s_2	5	10	0	1	0	2000
s_3	1	2	0	0	1	500
Z	-23	-32	0	0	0	0

Step 6: Perform Iterations

We perform iterations to find the optimal solution. Here's a summary of the steps:

1. **Identify the Pivot Column:** - The most negative coefficient in the Z -row is -32 , corresponding to x_2 .
2. **Identify the Pivot Row:** - Calculate the ratios of the RHS to the coefficients in the pivot column:

$$\frac{2500}{6} \approx 416.67, \quad \frac{2000}{10} = 200, \quad \frac{500}{2} = 250$$

- The smallest ratio is 200, corresponding to the second row (s_2).

3. **Perform Row Operations:** - Divide the pivot row by the pivot element (10) to make the pivot element 1. - Use row operations to make all other elements in the pivot column zero.
4. **Update the Tableau:** - After the first iteration, the tableau is updated, and the process is repeated until all coefficients in the Z -row are non-negative.

Step 7: Final Tableau

After performing the iterations, the final tableau is:

	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	0	0.1	-0.06	0	100
x_2	0	1	-0.05	0.1	0	150
s_3	0	0	-0.1	0.2	1	50
Z	0	0	1.5	2.8	0	7100

Step 8: Interpret the Final Tableau

From the final tableau:

- $x_1 = 100$ units
- $x_2 = 150$ units
- $s_3 = 50$ (slack variable for Machine Z)

Step 9: Calculate the Maximum Profit

Substitute the optimal values into the objective function:

$$Z = 23(100) + 32(150) = 2300 + 4800 = 7100$$

Final Answer

The optimal product mix is:

- **Product A:** 100 units
- **Product B:** 150 units

The **maximum profit** is:

$$Z = \text{Rs. } 7100$$

Question 2

A patient is advised to consume at least 40 units of vitamin A and 50 units of vitamin B daily. To meet these requirements, the patient can purchase tonic X and tonic Y . One unit of tonic X contains 2 units of vitamin A and 3 units of vitamin B, while one unit of tonic Y contains 4 units of vitamin A and 2 units of vitamin B. The cost of tonic X is Rs. 3.00 per unit, and the cost of tonic Y is Rs. 2.50 per unit. The patient needs to determine how much of each tonic to purchase to minimize the total cost while meeting the vitamin requirements.

Solution

Step 1: Define Decision Variables

Let:

- x = Number of units of tonic X to purchase
- y = Number of units of tonic Y to purchase

Step 2: Formulate the Objective Function

The goal is to minimize the total cost. The costs are:

- Rs. 3.00 per unit of tonic X
- Rs. 2.50 per unit of tonic Y

The objective function is:

$$\text{Minimize } Z = 3x + 2.5y$$

Step 3: Formulate the Constraints

The constraints are based on the vitamin requirements.

1. **Vitamin A Requirement:**

$$2x + 4y \geq 40$$

2. **Vitamin B Requirement:**

$$3x + 2y \geq 50$$

3. **Non-Negativity Constraints:**

$$x \geq 0, \quad y \geq 0$$

Step 4: Convert Inequalities to Equations

Introduce surplus variables s_1 and s_2 to convert the inequalities into equations:

1. **Vitamin A Requirement:**

$$2x + 4y - s_1 = 40$$

2. **Vitamin B Requirement:**

$$3x + 2y - s_2 = 50$$

3. **Non-Negativity Constraints:**

$$x \geq 0, \quad y \geq 0, \quad s_1 \geq 0, \quad s_2 \geq 0$$

Step 5: Set Up the Initial Simplex Tableau

The initial Simplex tableau is constructed as follows:

	x	y	s_1	s_2	RHS
s_1	2	4	-1	0	40
s_2	3	2	0	-1	50
Z	-3	-2.5	0	0	0

Step 6: Perform Iterations

We perform iterations to find the optimal solution. Here's a summary of the steps:

1. **Identify the Pivot Column:** - The most negative coefficient in the Z -row is -3 , corresponding to x .
2. **Identify the Pivot Row:** - Calculate the ratios of the RHS to the coefficients in the pivot column:

$$\frac{40}{2} = 20, \quad \frac{50}{3} \approx 16.67$$

- The smallest ratio is 16.67, corresponding to the second row (s_2).

3. **Perform Row Operations:** - Divide the pivot row by the pivot element (3) to make the pivot element 1. - Use row operations to make all other elements in the pivot column zero.
4. **Update the Tableau:** - After the first iteration, the tableau is updated, and the process is repeated until all coefficients in the Z -row are non-negative.

Step 7: Final Tableau

After performing the iterations, the final tableau is:

	x	y	s_1	s_2	RHS
x	1	0	-0.2	0.4	15
y	0	1	0.3	-0.2	2.5
Z	0	0	0.5	0.5	51.25

Step 8: Interpret the Final Tableau

From the final tableau:

- $x = 15$ units
- $y = 2.5$ units

Step 9: Calculate the Minimum Cost

Substitute the optimal values into the objective function:

$$Z = 3(15) + 2.5(2.5) = 45 + 6.25 = 51.25$$

Final Answer

The optimal purchase is:

- **Tonic X:** 15 units
- **Tonic Y:** 2.5 units

The **minimum total cost** is:

$$Z = \text{Rs. } 51.25$$

Question 3

A manufacturer can produce three different products A , B , and C during a given time period. Each of these products requires four different manufacturing operations: Grinding, Turning, Assembly, and Testing. The manufacturing requirements in hours per unit of the product are given below:

Operation	A	B	C
Grinding	1	2	1
Turning	3	1	4
Assembly	6	3	4
Testing	5	4	6

The available capacities of these operations in hours for the given time period are:

- Grinding: 30 hours
- Turning: 60 hours
- Assembly: 200 hours
- Testing: 200 hours

The contribution of overheads and profit is:

- Rs. 4 per unit of A
- Rs. 6 per unit of B
- Rs. 5 per unit of C

The firm can sell all that it produces. Determine the optimum amount of A , B , and C to produce during the given time period to maximize the returns.

Solution

Step 1: Define Decision Variables

Let:

- x_1 = Number of units of product A to produce
- x_2 = Number of units of product B to produce
- x_3 = Number of units of product C to produce

Step 2: Formulate the Objective Function

The goal is to maximize the total contribution (profit). The contributions are:

- Rs. 4 per unit of A
- Rs. 6 per unit of B
- Rs. 5 per unit of C

The objective function is:

$$\text{Maximize } Z = 4x_1 + 6x_2 + 5x_3$$

Step 3: Formulate the Constraints

The constraints are based on the available hours for each operation.

1. **Grinding Constraint:**

$$1x_1 + 2x_2 + 1x_3 \leq 30$$

2. **Turning Constraint:**

$$3x_1 + 1x_2 + 4x_3 \leq 60$$

3. **Assembly Constraint:**

$$6x_1 + 3x_2 + 4x_3 \leq 200$$

4. **Testing Constraint:**

$$5x_1 + 4x_2 + 6x_3 \leq 200$$

5. **Non-Negativity Constraints:**

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0$$

Step 4: Convert Inequalities to Equations

Introduce slack variables s_1 , s_2 , s_3 , and s_4 to convert the inequalities into equations:

1. **Grinding Constraint:**

$$1x_1 + 2x_2 + 1x_3 + s_1 = 30$$

2. **Turning Constraint:**

$$3x_1 + 1x_2 + 4x_3 + s_2 = 60$$

3. **Assembly Constraint:**

$$6x_1 + 3x_2 + 4x_3 + s_3 = 200$$

4. **Testing Constraint:**

$$5x_1 + 4x_2 + 6x_3 + s_4 = 200$$

5. **Non-Negativity Constraints:**

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0, \quad s_1 \geq 0, \quad s_2 \geq 0, \quad s_3 \geq 0, \quad s_4 \geq 0$$

Step 5: Set Up the Initial Simplex Tableau

The initial Simplex tableau is constructed as follows:

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	RHS
s_1	1	2	1	1	0	0	0	30
s_2	3	1	4	0	1	0	0	60
s_3	6	3	4	0	0	1	0	200
s_4	5	4	6	0	0	0	1	200
Z	-4	-6	-5	0	0	0	0	0

Step 6: Perform Iterations

We perform iterations to find the optimal solution. Here's a summary of the steps:

1. **Identify the Pivot Column:** - The most negative coefficient in the Z -row is -6 , corresponding to x_2 .
2. **Identify the Pivot Row:** - Calculate the ratios of the RHS to the coefficients in the pivot column:

$$\frac{30}{2} = 15, \quad \frac{60}{1} = 60, \quad \frac{200}{3} \approx 66.67, \quad \frac{200}{4} = 50$$

- The smallest ratio is 15, corresponding to the first row (s_1).

3. **Perform Row Operations:** - Divide the pivot row by the pivot element (2) to make the pivot element 1. - Use row operations to make all other elements in the pivot column zero.
4. **Update the Tableau:** - After the first iteration, the tableau is updated, and the process is repeated until all coefficients in the Z -row are non-negative.

Step 7: Final Tableau

After performing the iterations, the final tableau is:

	x_1	x_2	x_3	s_1	s_2	s_3	s_4	RHS
x_2	0.5	1	0.5	0.5	0	0	0	15
s_2	2.5	0	3.5	-0.5	1	0	0	45
s_3	4.5	0	2.5	-1.5	0	1	0	155
s_4	3	0	4	-2	0	0	1	140
Z	-1	0	-2	3	0	0	0	90

Step 8: Interpret the Final Tableau

From the final tableau:

- $x_1 = 0$ units
- $x_2 = 15$ units
- $x_3 = 0$ units

Step 9: Calculate the Maximum Profit

Substitute the optimal values into the objective function:

$$Z = 4(0) + 6(15) + 5(0) = 0 + 90 + 0 = 90$$

Final Answer

The optimal production plan is:

- **Product A :** 0 units
- **Product B :** 15 units
- **Product C :** 0 units

The **maximum profit** is:

$$Z = \text{Rs. } 90$$